



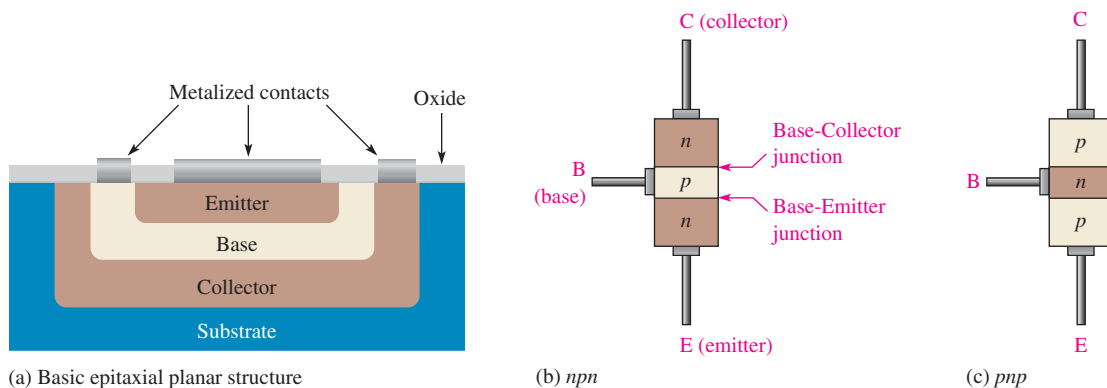
1 DC OPERATION OF BIPOLAR JUNCTION TRANSISTORS (BJTs)

A **transistor** is a semiconductor device that controls current between two terminals based on the current or voltage at a third terminal and is used for the amplification or switching of electrical signals. The basic structure of the bipolar junction transistor, BJT, determines its operating characteristics. DC bias is important to the operation of transistors in terms of setting up proper currents and voltages in a transistor circuit. Two important parameters are α_{DC} and β_{DC} .

After completing this section, you should be able to

- ◆ Describe the basic structure and operation of bipolar junction transistors
 - ◆ Explain the difference between *npn* and *pnp* transistors
 - ◆ Explain transistor biasing
 - ◆ Discuss transistor currents and their relationships
 - ◆ Define β_{DC}
 - ◆ Discuss the transistor voltages
 - ◆ Calculate dc bias currents and voltages in a basic transistor circuit

The **bipolar junction transistor (BJT)** is constructed with three doped semiconductor regions separated by two *pn* junctions, as shown in the epitaxial planar structure in Figure 1(a). The three regions are called **emitter**, **base**, and **collector**. Representations of the two types of bipolar transistors are shown in Figure 1(b) and (c). One type consists of two *n* regions separated by a *p* region (*npn*), and the other consists of two *p* regions separated by an *n* region (*pnp*).



▲ FIGURE 1

Basic construction of bipolar junction transistors.

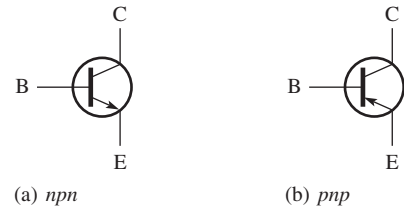
The *pn* junction joining the base region and the emitter region is called the *base-emitter junction*. The junction joining the base region and the collector region is called the *base-collector junction*, as indicated in Figure 1(b). A wire lead connects to each of the three regions, as shown. These leads are labeled E, B, and C for emitter, base, and collector, respectively. The base material is lightly doped and very narrow compared to the heavily doped emitter and collector materials.

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Figure 2 shows the schematic symbols for the *npn* and *pnp* bipolar transistors. Notice that the emitter terminal has an arrow. The term **bipolar** refers to the use of both holes and electrons as charge carriers in the transistor structure. Although both *pnp* and *npn* transistors are commonly used, the *npn* will be predominately used to illustrate concepts in this chapter.

► **FIGURE 2**

Transistor symbols.

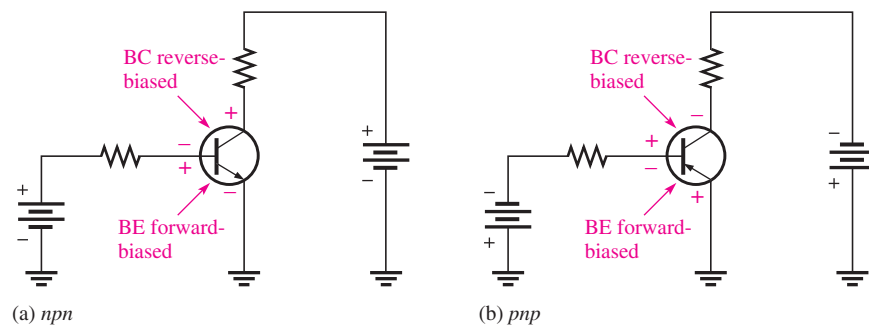


Transistor Biasing

In order for a transistor to operate properly as an amplifier, the two *pn* junctions must be correctly biased with external dc voltages. Figure 3 shows the proper bias arrangement for both *npn* and *pnp* transistors. Notice that in both cases the base-emitter (BE) junction is forward-biased and the base-collector (BC) junction is reverse-biased. This is called *forward-reverse bias*.

► **FIGURE 3**

Forward-reverse bias of a BJT.

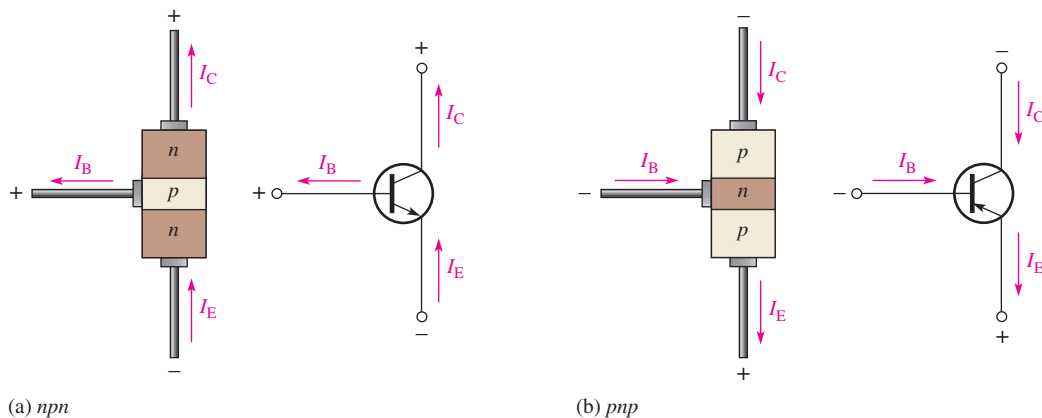


Transistor Currents

The directions of current in an *npn* and a *pnp* transistor are as shown in Figure 4(a) and 4(b), respectively. An examination of these diagrams shows that the emitter current is the sum of the collector and base currents, expressed as follows:

Equation 1

$$I_E = I_C + I_B$$



► **FIGURE 4**

Transistor currents.

As mentioned before, I_B is very small compared to I_E or I_C . The capital-letter subscripts indicate dc values.

These direct currents (emitter, base, and collector) are also related by two parameters: the dc **alpha** (α_{DC}), which is the ratio I_C/I_E and the dc **beta** (β_{DC}), which is the ratio I_C/I_B . β_{DC} is the direct current gain and is usually designated as h_{FE} on the transistor data sheets.

The collector current is equal to α_{DC} times the emitter current.

$$I_C = \alpha_{DC} I_E$$

Equation 2

where α_{DC} typically has a value between 0.950 and 0.995. Generally α_{DC} can be considered to be approximately 1 and therefore $I_C \cong I_E$.

The collector current is equal to the base current multiplied by β_{DC} .

$$I_C = \beta_{DC} I_B$$

Equation 3

where β_{DC} typically has a value between 20 and 300, depending on the type of transistor. Some specialized transistors can have much higher values.

Transistor Voltages

The three dc voltages for the biased transistor in Figure 5 are the emitter voltage (V_E), the collector voltage (V_C), and the base voltage (V_B). These voltages are with respect to ground. The collector voltage is equal to the dc supply voltage, V_{CC} , less the drop across R_C .

$$V_C = V_{CC} - I_C R_C$$

Equation 4

The base voltage is equal to the emitter voltage plus the base-emitter junction barrier potential (V_{BE}), which is about 0.7 V for a silicon transistor.

$$V_B = V_E + V_{BE}$$

Equation 5

In the configuration of Figure 5, the emitter is the common (grounded) terminal, so $V_E = 0$ V and $V_B = 0.7$ V.

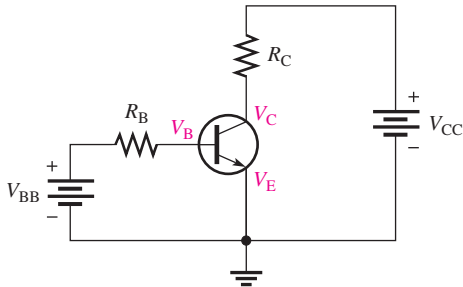


FIGURE 5

Bias voltages.

Voltage-Divider Bias

The two dc bias sources can be replaced by a single dc source and a resistive voltage divider. Although there are other bias methods, this one is the most widely used.

The voltage-divider bias configuration uses a single dc source to provide forward-bias to the transistor, as shown in Figure 6. Resistors R_1 and R_2 form a voltage divider that provides the base bias voltage. Resistor R_E allows the emitter to rise above ground potential.

The voltage divider provides a very small base current to the transistor compared to the bias current. For a properly designed circuit, the loading effect is negligible and can be ignored.

The input resistance at the transistor base depends on β_{DC} and is approximately

$$R_{IN} \cong \beta_{DC} R_E$$

Equation 6

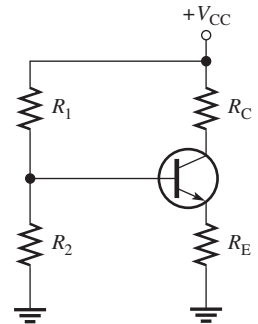


FIGURE 6

Voltage-divider bias.

Base Voltage

Using the voltage-divider formula, you get the following equation for the approximate base voltage for the circuit in Figure 6:

Equation 7

$$V_B \cong \left(\frac{R_2}{R_1 + R_2} \right) V_{CC}$$

Once you have determined the base voltage, you can determine the emitter voltage V_E (for an *npn* transistor).

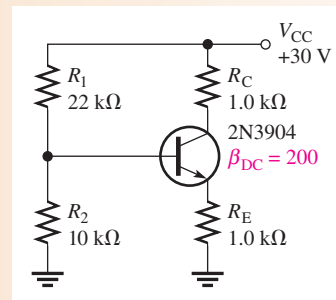
Equation 8

$$V_E = V_B - 0.7 \text{ V}$$

EXAMPLE 1

Determine V_B , V_E , V_C , V_{CE} , I_B , I_E , and I_C in Figure 7. The 2N3904 is a general-purpose transistor with a typical $\beta_{DC} = 200$.

► FIGURE 7



Solution The base voltage is approximately

$$V_B \cong \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \left(\frac{10 \text{ k}\Omega}{32 \text{ k}\Omega} \right) 30 \text{ V} = \mathbf{9.38 \text{ V}}$$

Therefore,

$$V_E = V_B - 0.7 \text{ V} = \mathbf{8.68 \text{ V}}$$

Now that you know V_E , you can find I_E by Ohm's law.

$$I_E = \frac{V_E}{R_E} = \frac{8.68 \text{ V}}{1.0 \text{ k}\Omega} = \mathbf{8.68 \text{ mA}}$$

Since α_{DC} is so close to 1 for most transistors, it is a good approximation to assume that $I_C \cong I_E$. Thus,

$$I_C \cong \mathbf{8.68 \text{ mA}}$$

Use $I_C = \beta_{DC} I_B$ and solve for I_B .

$$I_B = \frac{I_C}{\beta_{DC}} = \frac{8.68 \text{ mA}}{200} = \mathbf{43.4 \mu\text{A}}$$

Since you know I_C , you can find V_C .

$$V_C = V_{CC} - I_C R_C = 30 \text{ V} - (8.68 \text{ mA})(1.0 \text{ k}\Omega) = 30 \text{ V} - 8.68 \text{ V} = \mathbf{21.3 \text{ V}}$$

Since V_{CE} is the collector-to-emitter voltage, it is the difference of V_C and V_E .

$$V_{CE} = V_C - V_E = 21.3 \text{ V} - 8.68 \text{ V} = \mathbf{12.6 \text{ V}}$$

Related Problem*

Determine V_B , V_E , V_C , V_{CE} , I_B , I_E , and I_C if V_{CC} is changed to 15 V in Figure 7.

Open Multisim file E17-01; files are found at www.pearsonhighered.com/floyd/. Measure all of the transistor currents and the terminal voltages.

*Answers are at the end of the chapter.

SECTION 1 CHECKUP

Answers are at the end of the chapter.

1. Name the three regions of a bipolar junction transistor.
2. Define *forward-reverse bias*.
3. What is β_{DC} ?
4. If I_B is 10 μA and β_{DC} is 100, what is the collector current?
5. How many dc voltage sources are required for voltage-divider bias?
6. In Figure 7, if R_2 is 4.7 k Ω , what is the value of V_B ?

2 BJT CLASS A AMPLIFIERS

DC bias allows a transistor to operate as an amplifier. In a **class A amplifier**, the transistor conducts for the full cycle of the input signal (360°). Thus, a transistor can be used to produce a larger ac signal using a smaller ac signal as a “pattern.” In this section, we will discuss how a transistor acts as an amplifier and examine a basic amplifier.

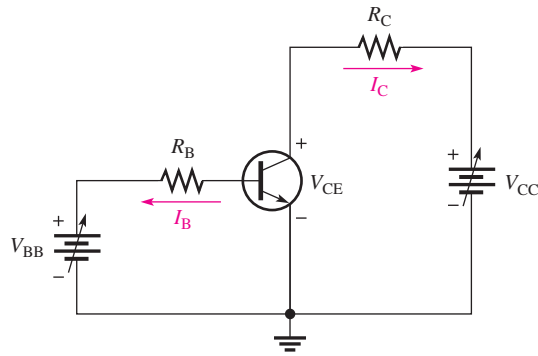
After completing this section, you should be able to

- ♦ **Explain the operation of a BJT class A amplifier**
 - ♦ Discuss the collector characteristic curves
 - ♦ Define *cutoff* and *saturation*
 - ♦ Analyze the dc load line operation of an amplifier
 - ♦ Discuss the significance of the Q-point
 - ♦ Explain the signal operation of an amplifier
 - ♦ Calculate the voltage, current, and power gains of common-emitter and common-collector amplifiers
 - ♦ Explain the use of coupling and bypass capacitors

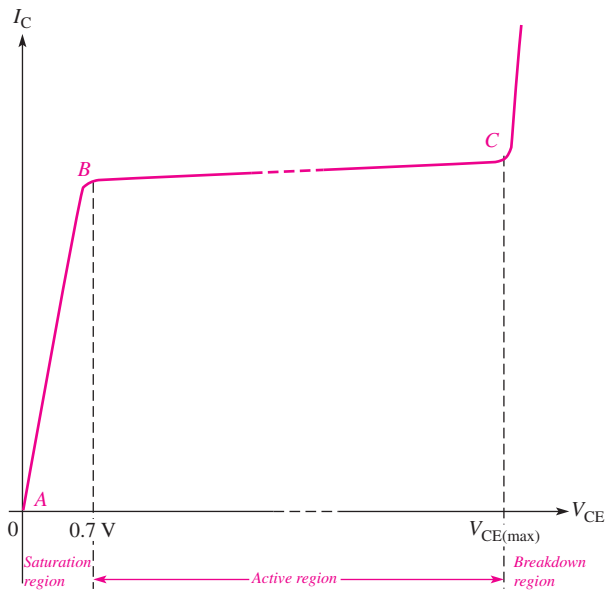
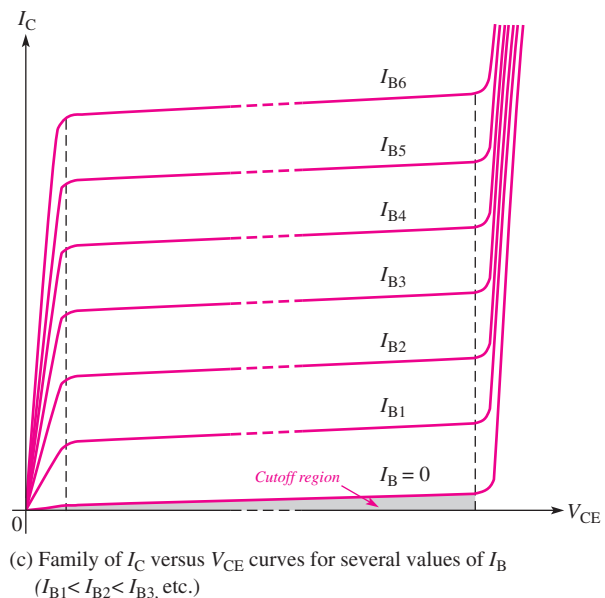
Let’s examine the factors important in the operation of a transistor class A **amplifier**. Generally, class A amplifiers are used in low-power applications where power levels are generally less than 1 W.

Collector Characteristic Curves

Using a circuit like that shown in Figure 8(a), you can generate a set of *collector characteristic curves* that show how the collector current, I_C , varies with the collector-to-emitter voltage, V_{CE} , for specified values of base current, I_B . Notice in the circuit diagram that both V_{BB} and V_{CC} are variable sources of voltage.



(a) Circuit


 (b) I_C versus V_{CE} curve for one value of I_B

 (c) Family of I_C versus V_{CE} curves for several values of I_B
 ($I_{B1} < I_{B2} < I_{B3}$, etc.)

▲ FIGURE 8

Collector characteristic curves.

Assume that V_{BB} is set to produce a certain value of I_B and V_{CC} is zero. For this condition, both the base-emitter junction and the base-collector junction are forward-biased because the base is at approximately 0.7 V while the emitter and the collector are at 0 V. Most of the base current is through the base-emitter junction because of the low impedance path to ground. When both junctions are forward-biased, the transistor is in the **saturation** region of its operation.

As V_{CC} is increased, V_{CE} increases gradually as the collector current increases. This is indicated by the portion of the characteristic curve between points A and B in Figure 8(b). I_C increases as V_{CC} is increased because V_{CE} remains less than 0.7 V due to the forward-biased base-collector junction.

Ideally, when V_{CE} exceeds 0.7 V, the base-collector junction becomes reverse-biased and the transistor goes into the **active** or **linear** region of its operation. Once the base-collector junction is reverse-biased, I_C levels off and remains essentially constant for a given value of I_B as V_{CE} continues to increase. Actually, I_C increases very slightly as V_{CE} increases due to widening of the base-collector depletion region. This results in fewer holes for recombination in the base region, which effectively causes a slight increase in β_{DC} . This is shown by the portion of the characteristic curve between points B and C in Figure 8(b). For this portion of the characteristic curve, the value of I_C is determined only by the relationship expressed as $I_C = \beta_{DC} I_B$.

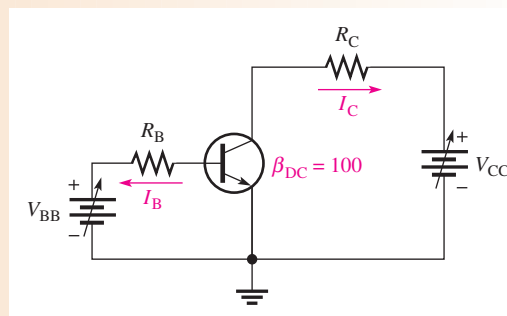
When V_{CE} reaches a sufficiently high voltage, the reverse-biased base-collector junction goes into breakdown; and the collector current increases rapidly as indicated by the part of the curve to the right of point C in Figure 8(b). A transistor should never be operated in this breakdown region.

A family of collector characteristic curves is produced when I_C versus V_{CE} is plotted for several values of I_B , as illustrated in Figure 8(c). When $I_B = 0$, the transistor is in the **cutoff** region although there is a very small collector leakage current as indicated. The amount of collector leakage current for $I_B = 0$ is exaggerated on the graph for purposes of illustration.

EXAMPLE 2

Draw the family of collector characteristic curves for the circuit in Figure 9 for $I_B = 5 \mu\text{A}$ to $25 \mu\text{A}$ in $5 \mu\text{A}$ increments. Assume that $\beta_{DC} = 100$.

► **FIGURE 9**

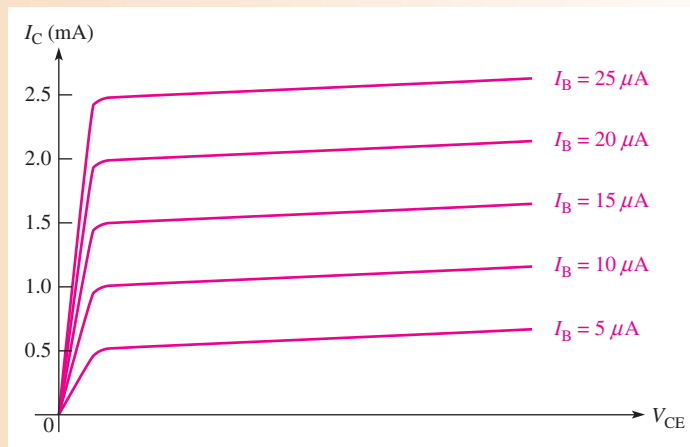


Solution Using the relationship $I_C = \beta_{DC} I_B$, you can calculate the ideal values of I_C and tabulate as in Table 1.

The resulting curves are plotted in Figure 10. Generally, characteristic curves have a slight upward slope as shown. This indicates that, for a given value of I_B , I_C increases slightly as V_{CE} increases, which implies β_{DC} is not exactly constant.

▼ **TABLE 1**

I_B	I_C
$5 \mu\text{A}$	0.5 mA
$10 \mu\text{A}$	1.0 mA
$15 \mu\text{A}$	1.5 mA
$20 \mu\text{A}$	2.0 mA
$25 \mu\text{A}$	2.5 mA



▲ **FIGURE 10**

Related Problem Where would the curve for $I_B = 0$ appear on the graph in Figure 10?

Cutoff and Saturation As previously mentioned, when $I_B = 0$, the transistor is in cutoff. Under this condition, there is a very small amount of collector leakage current, I_{CEO} , due mainly to thermally produced carriers. In cutoff, both the base-emitter and the base-collector junctions are reverse-biased.

Now let's consider the condition known as saturation. When the base current is increased, the collector current also increases and V_{CE} decreases as a result of more drop across R_C . When V_{CE} reaches a value called $V_{CE(sat)}$, the base-collector junction becomes forward-biased and I_C can increase no further even with a continued increase in I_B . At the point of saturation, the relation $I_C = \beta_{DC} I_B$ is no longer valid.

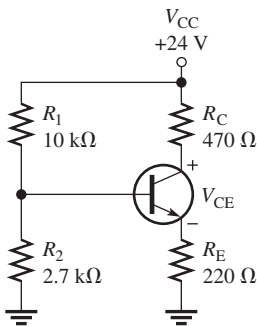
For a transistor, $V_{CE(sat)}$ occurs somewhere below the knee of the collector curves and is usually only a few tenths of a volt. Usually, $V_{CE(sat)}$ is assumed to be zero for analysis purposes, so a saturated transistor ideally acts as a short from collector to emitter.

Load Line Operation A straight line drawn on the collector curves between the cutoff and saturation points of the transistor is called the *load line*. Once set up, the transistor always operates along this line. Thus, any value of I_C and the corresponding V_{CE} will fall on this line. Notice that the load line is determined by the collector circuit resistance and V_{CC} and not by the transistor itself.

Now let's set up a dc load line for the circuit in Figure 11. First determine the cutoff point on the load line. When the transistor is in cutoff, there is essentially no collector current. Thus, the collector-emitter voltage, V_{CE} , is equal to V_{CC} . In this case, $V_{CE} = 24$ V.

Next, determine the saturation point on the load line. When the transistor is saturated, V_{CE} is approximately zero. (Actually, it is usually a few tenths of a volt, but zero is a good approximation.) Therefore, all the V_{CC} voltage is dropped across $R_C + R_E$. From this you can determine the saturation value of collector current, $I_{C(sat)}$. This value is the maximum value for I_C . You cannot possibly increase it further without changing V_{CC} , R_C , or R_E . In Figure 11, the value of $I_{C(sat)}$ is $V_{CC}/(R_C + R_E)$, which is 34.8 mA.

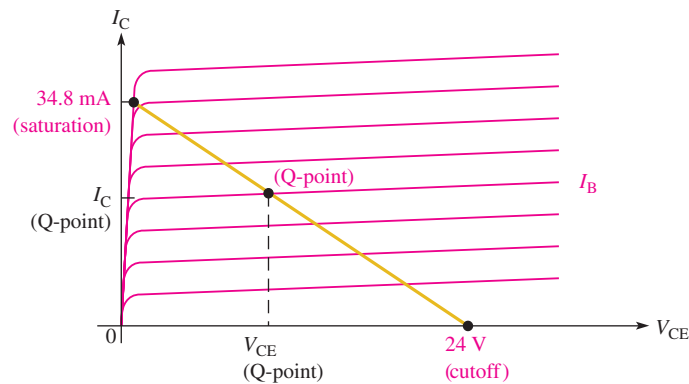
Finally, the cutoff and saturation points are plotted on the characteristic curves in Figure 12, and a straight line, which is the dc load line, is drawn between them.



▲ FIGURE 11

► FIGURE 12

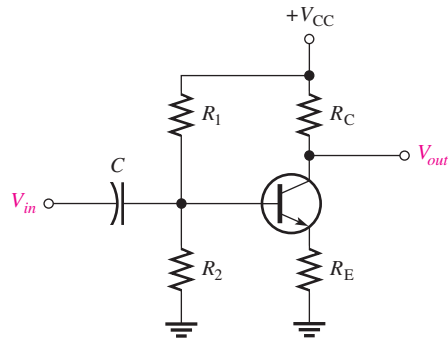
DC load line (gold) for the circuit in Figure 11.



Q-Point The base current, I_B , is established by the base bias. The point at which the base current curve intersects the dc load line is the *quiescent* or **Q-point** for the circuit. The coordinates of the Q-point are the values for I_C and V_{CE} at that point, as illustrated in Figure 12.

Signal (ac) Operation of an Amplifier

The circuit in Figure 13 produces an output signal with the same waveform as the input signal but with a greater amplitude. This increase is called **amplification**. The figure shows an input signal, V_{in} , capacitively coupled to the base. The collector voltage is the output signal, as indicated. The input signal voltage causes the base current to vary at the same frequency above and below its dc value. This variation in base current produces a corresponding variation in collector current. However, the variation in collector current is much larger than the variation in base current because of the current **gain** through the



◀ FIGURE 13

An amplifier with voltage-divider bias with capacitively coupled input signal. V_{in} and V_{out} are with respect to ground.

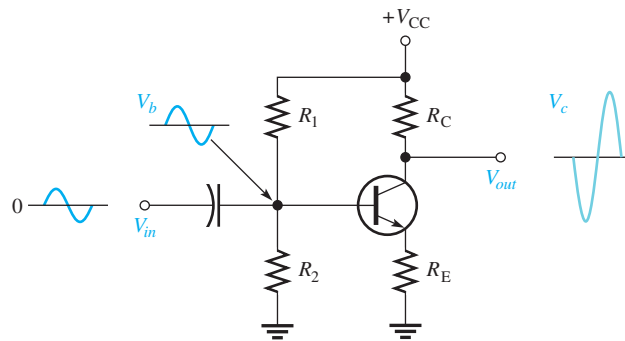
transistor. The ratio of the ac collector current (I_c) to the ac base current (I_b) is designated β_{ac} (the ac beta) or h_{fe} .

$$\beta_{ac} = \frac{I_c}{I_b}$$

Equation 9

The value of β_{ac} differs only slightly from that of β_{DC} for a given transistor and data sheets do not always show this parameter. Remember that lowercase subscripts distinguish ac currents and voltages from dc currents and voltages.

Signal Voltage Gain of an Amplifier Now let's take the amplifier in Figure 13 and examine its voltage gain with a signal input. The output voltage is the collector voltage. The variation in collector current produces a variation in the voltage across R_C and a resulting variation in the collector voltage, as shown in Figure 14.



◀ FIGURE 14

Voltage amplification.

As the collector current increases, the $I_c R_C$ drop increases. The increase produces a decrease in collector voltage because $V_c = V_{CC} - I_c R_C$. Likewise, as the collector current decreases, the $I_c R_C$ drop decreases and produces an increase in collector voltage. Therefore, there is a 180° phase difference between the collector current and the collector voltage. The base voltage and collector voltage are also 180° out of phase, as indicated in Figure 14. This 180° phase difference between input and output is called an *inversion*.

The voltage gain, A_v , of the amplifier is V_{out}/V_{in} , where V_{out} is the signal voltage at the collector and V_{in} is the signal voltage at the base. Because the base-emitter junction is forward-biased, the signal voltage at the emitter is approximately equal to the signal voltage at the base. Thus, since $V_b \cong V_e$, the gain is approximately $V_c/V_e = I_c R_C / I_e R_E$. Because the α_{ac} is close to 1, I_c and I_e are very close to the same value. Therefore, they cancel, giving the voltage gain formula for the amplifier in Figure 14:

$$A_v \cong \frac{R_C}{R_E}$$

Equation 10

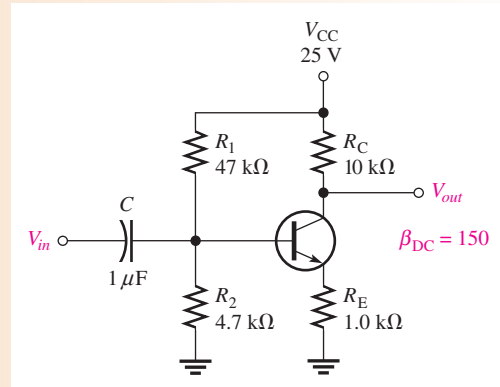
A negative sign on A_v is often used to indicate the inversion between input and output.

EXAMPLE 3

In Figure 15, a signal voltage of 50 mV rms is applied to the base.

- Determine the output signal voltage for the amplifier.
- Find the dc collector voltage on which the output signal voltage is riding.
- Draw the output waveform.

► **FIGURE 15**



Solution (a) The signal voltage gain is

$$A_v \cong \frac{R_C}{R_E} = \frac{10 \text{ k}\Omega}{1.0 \text{ k}\Omega} = 10$$

The output signal voltage is the input signal voltage times the voltage gain.

$$V_{out} = A_v V_{in} = (10)(50 \text{ mV}) = \mathbf{500 \text{ mV rms}}$$

(b) Next find the dc collector voltage.

$$V_B \cong \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \left(\frac{4.7 \text{ k}\Omega}{51.7 \text{ k}\Omega} \right) 25 \text{ V} = 2.27 \text{ V}$$

$$I_C \cong I_E = \frac{V_E}{R_E} = \frac{V_B - 0.7 \text{ V}}{1.0 \text{ k}\Omega} = 1.57 \text{ mA}$$

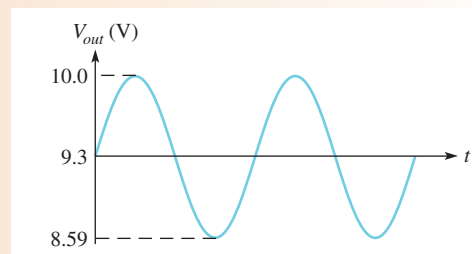
$$V_C = V_{CC} - I_C R_C = 25 \text{ V} - (1.57 \text{ mA})(10 \text{ k}\Omega) = \mathbf{9.3 \text{ V}}$$

This value is the dc level of the output. The peak value of the output signal is

$$V_p = 1.414(500 \text{ mV}) = 707 \text{ mV}$$

(c) Figure 16 shows the output waveform with the signal voltage riding on the 9.3 V dc level. Remember, the output waveform is inverted compared to the input.

► **FIGURE 16**

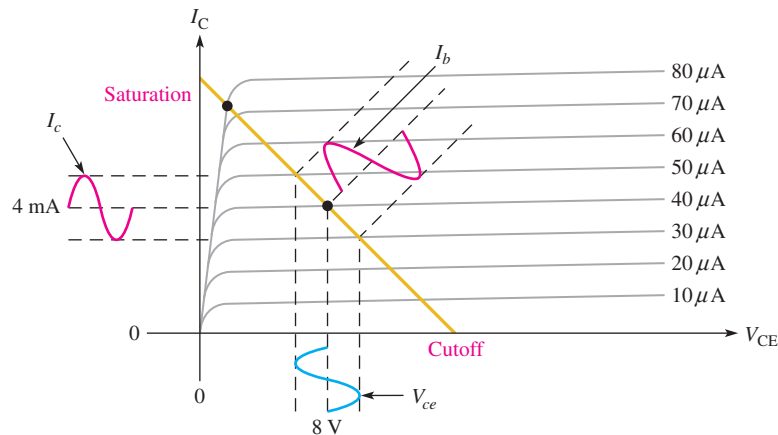


Related Problem If R_C is changed to $12\text{ k}\Omega$ in Figure 15, what is the output waveform?



Open Multisim file E17-03. Measure the output signal voltage and the dc level with the oscilloscope. Compare with the waveform in Figure 16.

Signal Operation on the Load Line You can obtain a graphical picture of an amplifier's operation by showing an example of signal variations on a set of collector curves with a load line, as shown in Figure 17. Let's assume that the dc Q-point values are as follows: $I_B = 40\text{ }\mu\text{A}$, $I_C = 4\text{ mA}$, and $V_{CE} = 8\text{ V}$. The input signal varies the base current from a maximum of $50\text{ }\mu\text{A}$ to a minimum of $30\text{ }\mu\text{A}$. The resulting variations in collector current and V_{CE} are shown on the graph. The operation is linear as long as the variations do not reach cutoff at the lower end of the load line or saturation at the upper end. An amplifier that operates entirely its linear region is a class A amplifier.

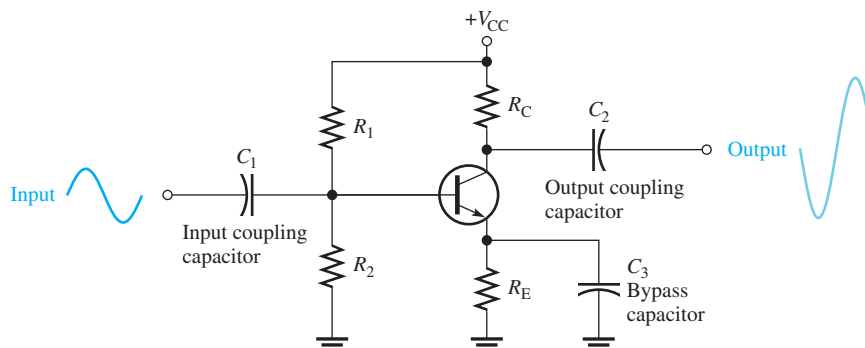


◀ **FIGURE 17**

Signal operation on the load line.

The Common-Emitter Amplifier

Figure 18 shows a typical **common-emitter (CE)** amplifier. The one shown has voltage-divider bias, although other types of bias methods are possible. C_1 and C_2 are coupling capacitors used to pass the signal into and out of the amplifier such that the source or load will not affect the dc bias voltages. C_3 is a bypass capacitor that shorts the emitter signal voltage (ac) to ground without disturbing the dc emitter voltage. Because of the bypass capacitor, the emitter is at signal (ac) ground (but not dc ground), thus making the circuit a common-emitter amplifier. The bypass capacitor increases the signal voltage gain. (The reason why this increase occurs is discussed next.) Notice that the input signal is applied to the



◀ **FIGURE 18**

Typical common-emitter (CE) amplifier.

base, and the output signal is taken from the collector. All capacitors are assumed to have a reactance of approximately zero at the signal frequency.

The Bypass Capacitor Increases Voltage Gain The bypass capacitor shorts the signal around the emitter resistor, R_E , in order to increase the voltage gain. To understand why, let's consider the amplifier without the bypass capacitor and see what the voltage is. Imagine the CE amplifier in Figure 18 with the bypass capacitor removed.

As before, lowercase italic subscripts indicate signal (ac) voltages and signal (alternating) currents. The **voltage gain** of the amplifier is V_{out}/V_{in} . The output signal voltage is

$$V_{out} = I_c R_C$$

The signal voltage at the base is approximately equal to

$$V_b \cong V_{in} \cong I_e(r_e + R_E)$$

where r_e is the internal emitter resistance of the transistor (not shown in the schematic). The voltage gain, A_v , can now be expressed as

$$A_v = \frac{V_{out}}{V_{in}} = \frac{I_c R_C}{I_e(r_e + R_E)}$$

Since $I_c \cong I_e$, the currents cancel and the gain is the ratio of the resistances.

Equation 11

$$A_v = \frac{R_C}{r_e + R_E}$$

Keep in mind that this formula is for the amplifier without the bypass capacitor. If R_E is much greater than r_e , then $A_v \cong R_C/R_E$ as given earlier in Equation 10.

If the bypass capacitor is connected across R_E , it effectively shorts the signal to ground, leaving only r_e in the emitter. Thus, the voltage gain of the CE amplifier with the bypass capacitor shorting R_E is

Equation 12

$$A_v = \frac{R_C}{r_e}$$

The transistor parameter r_e is important because it determines the voltage gain of a CE amplifier in conjunction with R_C . A formula for estimating r_e is given without derivation in the following equation:

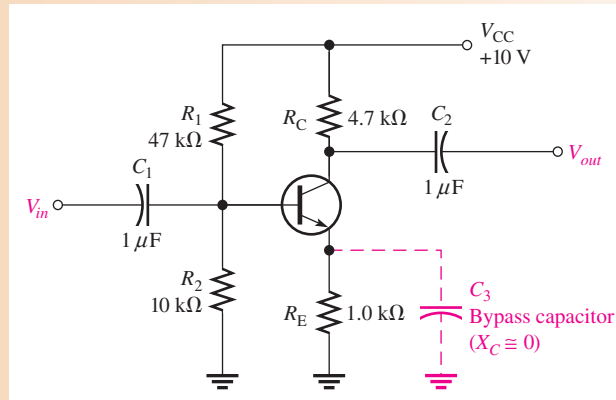
Equation 13

$$r_e \cong \frac{25 \text{ mV}}{I_E}$$

EXAMPLE 4

Determine the voltage gain of the amplifier in Figure 19 both with and without a bypass capacitor in normal and in decibel form.

► FIGURE 19



Solution First, determine r_e . To do so, you need to find I_E . Thus,

$$V_B \cong \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \left(\frac{10 \text{ k}\Omega}{47 \text{ k}\Omega + 10 \text{ k}\Omega} \right) 10 \text{ V} = 1.75 \text{ V}$$

$$V_E = V_B - 0.7 \text{ V} = 1.05 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.05 \text{ V}}{1.0 \text{ k}\Omega} = 1.05 \text{ mA}$$

$$r_e \cong \frac{25 \text{ mV}}{I_E} = \frac{25 \text{ mV}}{1.05 \text{ mA}} = 23.8 \Omega$$

The voltage gain without a bypass capacitor is

$$A_v = \frac{R_C}{r_e + R_E} = \frac{4.7 \text{ k}\Omega}{1023.8 \Omega} = \mathbf{4.59}$$

In decibel form,

$$A_v = 20 \log(4.59) = \mathbf{13.24 \text{ dB}}$$

The voltage gain with the bypass capacitor installed is

$$A_v = \frac{R_C}{r_e} = \frac{4.7 \text{ k}\Omega}{23.8 \Omega} = \mathbf{197}$$

As you can see, the voltage gain is greatly increased by the addition of the bypass capacitor. In terms of decibels (dB), the voltage gain is

$$A_v = 20 \log(197) = \mathbf{45.9 \text{ dB}}$$

Related Problem What is the voltage gain with the bypass capacitor if $R_C = 5.6 \text{ k}\Omega$?



Open Multisim file E17-04. Measure the voltage gain of the amplifier without a bypass capacitor. Connect a $10 \mu\text{F}$ bypass capacitor and measure the gain. Compare both measurements to the calculated values.

Phase Inversion As we discussed before, the output voltage at the collector is 180° out of phase with the input voltage at the base. Therefore, the CE amplifier is characterized by a phase inversion between the input and the output. This inversion is sometimes indicated by a negative voltage gain.

AC Input Resistance The dc input resistance (R_{IN}), viewed from the base of the transistor, was given in Equation 6. The input resistance “seen” by the signal at the base is derived in a similar manner when the emitter resistor is bypassed to ground.

$$\begin{aligned} R_{in} &= \frac{V_b}{I_b} \\ V_b &= I_e r_e \\ I_e &\cong \beta_{ac} I_b \\ R_{in} &\cong \frac{\beta_{ac} I_b r_e}{I_b} \end{aligned}$$

The I_b terms cancel, leaving

$$R_{in} \cong \beta_{ac} r_e$$

Equation 14

If the emitter resistor (R_E) is not bypassed, the emitter resistance must be added to r_e first.

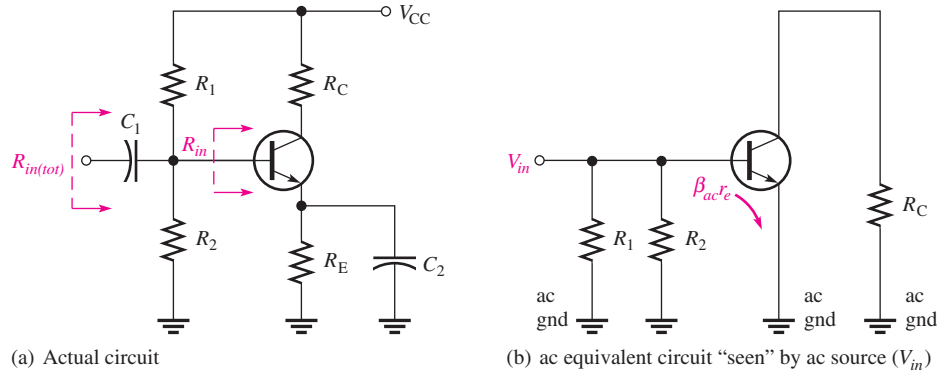
Total Input Resistance of a CE Amplifier Viewed from the base, R_{in} is the ac resistance. The actual resistance “seen” by the source includes that of bias resistors. We will now develop an expression for the total input resistance. The concept of ac ground was mentioned earlier. At this point it needs some additional explanation because it is important in the development of the formula for total input resistance, $R_{in(tot)}$.

You have already seen that the bypass capacitor effectively makes the emitter appear as ground to the ac signal because the reactance of the capacitor is nearly zero at the signal frequency. Of course, to a dc signal the capacitor looks like an open and thus does not affect the dc emitter voltage.

In addition to “seeing” ground through the bypass capacitor, the signal also “sees” ground through the dc supply voltage source, V_{CC} . It does so because there is zero signal voltage at the V_{CC} terminal. Thus, the $+V_{CC}$ terminal effectively acts as ac ground. As a result, the two bias resistors, R_1 and R_2 , appear in parallel to the ac input because one end of R_2 goes to actual ground and one end of R_1 goes to ac ground (V_{CC} terminal). Also, R_{in} at the base appears in parallel with $R_1 \parallel R_2$. This situation is illustrated in Figure 20. The collector resistor, R_C , does not affect the input resistance because the collector-base junction is reverse-biased.

► **FIGURE 20**

Total input resistance.



The expression for the total input resistance to the CE amplifier as “seen” by the ac source is

Equation 15

$$R_{in(tot)} = R_1 \parallel R_2 \parallel R_{in}$$

R_C has no effect because of the reverse-biased, base-collector junction.

Current Gain The signal **current gain** of a CE amplifier is

Equation 16

$$A_i = \frac{I_c}{I_s}$$

where I_s is the source current and is calculated by $V_{in}/R_{in(tot)}$.

Power Gain The **power gain** of a CE amplifier is the product of the voltage gain and the current gain.

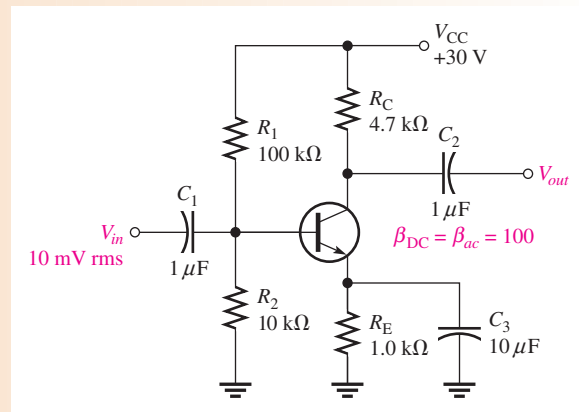
Equation 17

$$A_p = A_v A_i$$

EXAMPLE 5

Determine the voltage gain, current gain, and power gain for the CE amplifier in Figure 21. $\beta_{DC} = \beta_{ac} = 100$. Also, express the voltage and power gains in decibels.

► FIGURE 21



Solution First find r_e . To do so, you must find I_E . Begin by calculating V_B .

$$V_B \cong \left(\frac{R_2}{R_1 + R_2} \right) V_{CC} = \left(\frac{10 \text{ k}\Omega}{110 \text{ k}\Omega} \right) 30 \text{ V} = 2.73 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{V_B - 0.7 \text{ V}}{R_E} = \frac{2.03 \text{ V}}{1.0 \text{ k}\Omega} = 2.03 \text{ mA}$$

$$r_e \cong \frac{25 \text{ mV}}{I_E} = \frac{25 \text{ mV}}{2.03 \text{ mA}} = 12.3 \Omega$$

The ac voltage gain is

$$A_v \cong \frac{R_C}{r_e} = \frac{4.7 \text{ k}\Omega}{12.3 \Omega} = \mathbf{382}$$

Determine the signal current gain by first finding $R_{in(tot)}$ to get I_s .

$$R_{in(tot)} = R_1 \parallel R_2 \parallel \beta_{ac} r_e = 100 \text{ k}\Omega \parallel 10 \text{ k}\Omega \parallel 1.23 \text{ k}\Omega = 1.08 \text{ k}\Omega$$

$$I_s = \frac{V_{in}}{R_{in(tot)}} = \frac{10 \text{ mV}}{1.08 \text{ k}\Omega} = 9.26 \mu\text{A}$$

Next determine I_c .

$$I_c = \frac{V_{out}}{R_C} = \frac{A_v V_{in}}{R_C} = \frac{(382)(10 \text{ mV})}{4.7 \text{ k}\Omega} = \mathbf{813 \mu\text{A}}$$

$$A_i = \frac{I_c}{I_s} = \frac{813 \mu\text{A}}{9.26 \mu\text{A}} = \mathbf{87.8}$$

The power gain is

$$A_p = A_v A_i = (382)(87.8) = \mathbf{33,540}$$

The voltage gain and the power gain in decibels are as follows:

$$A_v = 20 \log(382) = \mathbf{51.6 \text{ dB}}$$

$$A_p = 10 \log(33,540) = \mathbf{45.3 \text{ dB}}$$

Related Problem Determine A_v in this example if C_3 is removed.



Open Multisim file E17-05. Measure the voltage gain and compare with the calculated value.

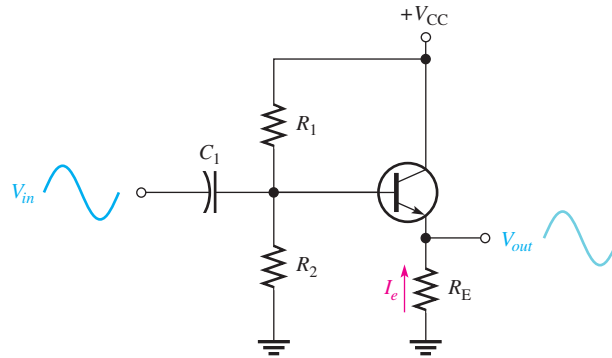
The Common-Collector Amplifier

The common-collector (CC) amplifier, commonly referred to as an emitter-follower, is another type of BJT amplifier configuration. The input is applied to the base and the output is at the emitter. There is no collector resistor. The voltage gain of a CC amplifier is approximately 1, but the current gain is always greater than 1.

Figure 22 shows a **common-collector (CC)** circuit with a voltage-divider bias. Notice that the input is applied to the base and the output is taken from the emitter.

► **FIGURE 22**

Typical emitter-follower (common-collector, CC) amplifier.



Voltage Gain As in all amplifiers, the voltage gain in a CC amplifier is $A_v = V_{out}/V_{in}$. For the emitter-follower, V_{out} is $I_e R_E$, and V_{in} is $I_e(r_e + R_E)$. Therefore, the gain is $I_e R_E / I_e(r_e + R_E)$. The currents cancel, and the gain expression simplifies to

Equation 18

$$A_v = \frac{R_E}{r_e + R_E}$$

It is important to notice here that *the gain is always less than 1*. Because r_e is normally much less than R_E , then a good approximation is $A_v = 1$.

Since the output voltage is the emitter voltage, it is in phase with the base or the input voltage. As a result, and because the voltage gain is close to 1, the output voltage follows the input voltage—thus the common-collector amplifier is also known as the **emitter-follower**.

Input Resistance The emitter-follower is characterized by a high input resistance, which makes it a very useful circuit. Because of the high input resistance, the emitter-follower can be used as a buffer to minimize loading effects when one circuit is driving another.

The derivation of the input resistance viewed from the base is similar to that for the CE amplifier. In this case, however, the emitter resistor is not bypassed.

$$R_{in} = \frac{V_b}{I_b} = \frac{I_e(r_e + R_E)}{I_b} \cong \frac{\beta_{ac} I_b(r_e + R_E)}{I_b} = \beta_{ac}(r_e + R_E)$$

If R_E is at least ten times larger than r_e , then the input resistance at the base is

Equation 19

$$R_{in} \cong \beta_{ac} R_E$$

In Figure 22, the bias resistors appear to the input signal to be in parallel with R_{in} , just as in the voltage-divider biased CE amplifier. The total ac input resistance is

Equation 20

$$R_{in(tot)} = R_1 \parallel R_2 \parallel R_{in}$$

Current Gain The signal current gain for the emitter-follower is I_e/I_s where I_s is the signal current and can be calculated as $V_s/R_{in(tot)}$. If the bias resistors are large enough to be

neglected so that $I_s = I_b$, then the current gain of the amplifier is equal to the current gain of the transistor, β_{ac} . Of course, the same was also true for the CE amplifier. β_{ac} is the maximum achievable current gain in both types of amplifiers.

$$A_i = \frac{I_e}{I_s} \quad \text{Equation 21}$$

Since $I_e = V_{out}/R_E$ and $I_s = V_{in}/R_{in(tot)}$, A_i can also be expressed as $R_{in(tot)}/R_E$, as will be shown in Example 6.

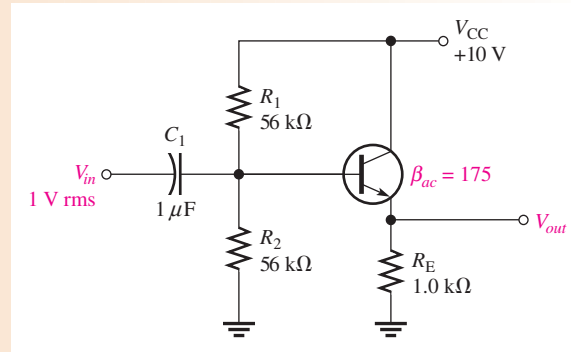
Power Gain The power gain is the product of the voltage gain and the current gain. For the emitter-follower, the power gain is approximately equal to the current gain because the voltage gain is approximately 1.

$$A_p \cong A_i \quad \text{Equation 22}$$

EXAMPLE 6

Determine the input resistance of the emitter-follower in Figure 23. Also find the voltage gain, current gain, and power gain.

► FIGURE 23



Solution The approximate input resistance viewed from the base is

$$R_{in} \cong \beta_{ac} R_E = (175)(1.0 \text{ k}\Omega) = 175 \text{ k}\Omega$$

The total input resistance is

$$R_{in(tot)} = R_1 \parallel R_2 \parallel R_{in} = 56 \text{ k}\Omega \parallel 56 \text{ k}\Omega \parallel 175 \text{ k}\Omega = \mathbf{24.1 \text{ k}\Omega}$$

The voltage gain is, neglecting r_e ,

$$A_v \cong 1$$

The current gain is

$$\begin{aligned} A_i &= \frac{I_e}{I_s} \\ I_e &= \frac{V_{out}}{R_E} \\ I_s &= \frac{V_{in}}{R_{in(tot)}} \\ A_i &= \frac{V_{out}/R_E}{V_{in}/R_{in(tot)}} \end{aligned}$$

Since $V_{in} \cong V_{out}$,

$$A_i = \frac{R_{in(tot)}}{R_E} = \frac{24.1 \text{ k}\Omega}{1.0 \text{ k}\Omega} = \mathbf{24.1}$$

The power gain is

$$A_p \cong A_i = \mathbf{24.1}$$

Related Problem If R_E in Figure 23 is decreased to 820Ω , what is the power gain?



Open Multisim file E17-06. Verify that the voltage gain is close to 1.

In addition to the common-emitter (CE) and the common-collector amplifier (CC) configurations, there is a third configuration called the **common-base (CB)**. In the CB configuration, the base is the common (grounded) terminal, the input signal is applied to the emitter, and the output signal is taken off the collector. The CB exhibits voltage gains greater than 1, similar to the CE configuration, but the current gain is always less than 1. The CB configuration is not widely used in amplifier applications.

SECTION 2 CHECKUP

1. What does the term *amplification* mean?
2. What is the Q-point?
3. A common-emitter circuit has $R_C = 47 \text{ k}\Omega$ and $R_E = 2.2 \text{ k}\Omega$. What is the approximate voltage gain?
4. What is the purpose of the bypass capacitor in a CE amplifier?
5. How is the voltage gain of a CE amplifier determined?
6. If A_v is 50 and A_i is 94, what is the power gain of a CE amplifier?
7. What is another name for a common-collector amplifier?
8. What is the ideal maximum voltage gain of a CC amplifier?

3 BJT CLASS B AMPLIFIERS

When an amplifier is biased such that it operates in the linear region for 180° of the input cycle and is in cutoff for 180° , it is a **class B amplifier**. The primary advantage of a class B amplifier over a class A is that the class B is more efficient; you can get more output power for a given amount of input power. A disadvantage of class B amplifiers is that it is more difficult to implement the circuit in order to get a linear reproduction of the input waveform. As you will see in this section, the term **push-pull** refers to a common type of class B amplifier circuit in which the input wave shape is approximately reproduced at the output.

After completing this section, you should be able to

- ♦ **Analyze class B amplifiers**
 - ♦ Explain class B operation
 - ♦ Discuss the meaning of push-pull amplifier operation